COMS4060A/7056A: Assignment # 2

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Introduction

This assignment is based on content covering geospatial and time series data. You will be required to perform some basic data cleaning and exploration techniques on a prescribed dataset. The aim is to explore the dataset and make observations. You are required to provide a Jupyter notebook and a PDF for all your reasoning. You must discuss all your answers on the PDF with the plots attached. You must provide one Jupyter notebook that clearly mentions the question number you are solving. You are free to use any library/framework in Python.

Usefult Python Packages

You might find the following packages particularly useful, but you are free to use any other package:

- folium
- fiona
- shapely
- geopandas
- sklearn.cluster.DBSCAN
- sklearn.decomposition.PCA
- sklearn.manifold.TSNE
- umap-learn (imported as `umap`)

The following method might be useful as well (note this is for numpy, but you can create a standard Python version if you wish):

```
def haversine_np(lon1, lat1, lon2, lat2):
    """
    Calculate the great circle distance between two points
    on the earth (specified in decimal degrees)
    All args must be of equal length.
    """
    lon1, lat1, lon2, lat2 = map(np.radians, [lon1, lat1, lon2, lat2])
    dlon = lon2 - lon1
    dlat = lat2 - lat1
    a = np.sin(dlat/2.0)**2 + np.cos(lat1) * np.cos(lat2) * np.sin(dlon/2.0)**2
    c = 2 * np.arcsin(np.sqrt(a))
    km = 6367 * c
    return km
```

NYC Taxi Data

You will analyse a public dataset from Uber available on Kaggle, available here: `https://www.kaggle.com/c/nyc-taxi-trip-duration`. It is also available on Moodle for download.

Your primary dataset is one released by the NYC Taxi and Limousine Commission (TLC), which includes pickup time, geo-coordinates, number of passengers, and several other variables for 1.5 million trips between 2016-01-01 and 2016-06-30. Note that for this analysis, just use the training sample.

- id - a unique identifier for each trip
- vendor_id - a code indicating the provider associated with the trip record
- pickup_datetime - date and time when the meter was engaged
- dropoff_datetime - date and time when the meter was disengaged
- passenger_count - the number of passengers in the vehicle (driver entered value)
- pickup_longitude - the longitude where the meter was engaged
- pickup_latitude - the latitude where the meter was engaged
- dropoff_longitude - the longitude where the meter was disengaged
- dropoff_latitude - the latitude where the meter was disengaged
- store_and_fwd_flag - This flag indicates whether the trip record was held in vehicle memory before sending to the vendor because the vehicle did not have a connection to the server - Y=store and forward; N=not a store and forward trip
- trip_duration - duration of the trip in seconds

There is more up-to-date data available from TLC, but the datasets are large (10GB+ per year since 2018). They do include additional fields, however. For this exercise, you can assume the centre of New York City is at the long/lat coordinates (40.716662, -74.009899).

1 Data Cleaning [3 marks]

There are several outliers in the data. Identify these and give justification for why you can remove them from the analysis. (Hint: look at trip duration, speed, distance, etc). [3 marks]

2 Feature generation [3 marks]

This can be done before or after the data cleaning step. Generate additional columns for at least these features (but you're welcome to add more!):

- Distance of trip
- Day of week
- Average speed of trip

[3 marks]

3 Time-based [14 marks]

Assume pickup time unless otherwise specified.

1. Which day of the week is the most popular? Show plots to motivate your answer. [2 marks]
2. What hour of the day is the most popular on each day? Plot a distribution of the hours and make observations and give possible suggestions for why the data looks like it does. [3 marks]
3. Investigate the differences between weekdays and weekends. What would account for this? [2 marks]
4. Look at how these patterns change on the major holidays (do they change?). Look at the following: St. Patrick's Day, Easter, Memorial Day, Valentine's Day, Martin Luther King Day. Make sure you use the correct dates for these for the relevant year. [5 marks]
5. How does the average speed of trips change throughout the day? What time of day are trips fastest? Show plots to motivate your answer. [2 marks]

4 Location clusters [12 marks]

4.1 Heatmaps

Produce a heatmap of all of the trip pickups over (do not do a scatter plot... there are 1.5 million data points and this will almost certainly crash your computer):

1. weekdays and weekends,
2. morning and evening (choose reasonable hours).

Comment on any findings you make. [4 marks]

4.2 Hotspots

If you were a taxi driver wanting to plan your evenings so that you could get the most trips, you would want to know where the popular areas are. Looking at the time periods 23:00 on a Friday evening to 02:00 on a Saturday morning, and between 17:00 and 20:00 on a Thursday, find hotspot locations (areas where there are a large number of trips happening). If you were to use k-means, you would define the number of clusters. However, here the number of clusters is not at all clear. DBSCAN (available in sklearn) determines this for you, and works well on spatial data. DBSCAN has two configurable parameters: ϵ - the maximum distance between any two points, and the minimum number of samples to determine a cluster. Your hotspot location might be defined as at least 15 pickups in that location in an hour, and locations might be required to be within 50 or 100 metres from each other (motivate your choice of parameters). Using DBSCAN, identify clusters and plot these on a map. How many clusters did you find? [8 marks]

The following code might be useful (but edit it to suit your own needs):

```
# calculate epsilon parameter using the user defined distance
kms_per_radian = 6371.0088
# The epsilon parameter is the max distance that points can be
# from each other to be
considered a cluster.
epsilon = max_distance / kms_per_radian
db = DBSCAN(eps=epsilon,
            min_samples=min_cars,
            algorithm="ball_tree",
            metric="haversine").fit(np.radians(coords))
```

5 Airports [9 marks]

Find out how long it takes, on average, to travel to JFK airport from the Empire State Building. Produce a plot showing the travel time by time of day. How

does this compare with Newark Airport? Assume the following coordinates for the centre point of the locations (long, lat):

- JFK Airport: (40.647929, -73.777813)
- Empire State Building: (40.756724, -73.983806)
- Newark Airport: (40.689442, -74.173242)

Use a reasonable (and motivate!) radius around these locations when determining if a GPS coordinate is at that location. [9 marks]

6 Dimensionality Reduction [22 marks]

Each trip can be described by many features (location, time, distance, speed, etc.), which makes it hard to see the structure of the data directly. PCA is a linear technique that projects the data onto the directions of maximum variance, while t-SNE and UMAP are non-linear techniques that project high-dimensional data into two dimensions while trying to preserve neighbourhood relationships. In this question you will use all three to explore whether there are natural "types" of taxi trips.

t-SNE does not scale to 1.5 million points, so work with a random sample of 20 000 trips from your *cleaned* dataset (fix the random seed so that your results are reproducible).

1. Build a feature matrix for the sampled trips. Include at least the pickup and dropoff coordinates, trip distance, trip duration, average speed, and pickup hour and day of week. Encode the hour of the day cyclically (e.g. using $\sin(2\pi h/24)$ and $\cos(2\pi h/24)$) and scale your features. Justify your choice of features, encoding, and scaling. [2 marks]

2. Apply PCA to the feature matrix. Plot the explained variance ratio of each component (scree plot) together with the cumulative explained variance, and state how many components are needed to explain 90% of the variance. Plot the trips on the first two principal components, coloured by (i) pickup hour, (ii) weekday vs. weekend, and (iii) trip distance. Inspect the loadings of the first two components and interpret what each component represents. [4 marks]

3. Apply t-SNE to the feature matrix using at least three different perplexity values (e.g. 5, 30, and 100). Colour the embeddings by (i) pickup hour, (ii) weekday vs. weekend, and (iii) trip distance. How does the perplexity affect the structure you observe? [3 marks]

4. Apply UMAP to the same feature matrix, varying `n_neighbors` (e.g. 5, 15, 50) and `min_dist` (e.g. 0.0 and 0.5). Produce the same coloured plots as above and discuss the effect of each parameter. [3 marks]

5. Cluster the trips in each of the three reduced spaces using a density-based method (DBSCAN or HDBSCAN, both available in `sklearn.cluster`):

   - for PCA, cluster on the number of components you found explains 90% of the variance, and visualise the result on the first two components;
   - for t-SNE and UMAP, cluster on your best 2D embedding from the previous parts.

   Motivate your clustering parameters (e.g. use a k-distance plot to choose ϵ for DBSCAN) and report the number of clusters and the fraction of trips labelled as noise for each method. For each cluster, produce a profile of the original (unscaled) features, such as the median distance, duration, speed, and pickup hour and the weekday/weekend split, and plot its pickup locations on a map. Use these profiles to give each cluster a descriptive name (e.g. "late-night airport runs" or "short midtown weekday hops"). [6 marks]

6. Compare PCA, t-SNE, and UMAP. Place your best embedding from each method side by side, using the same colouring, and discuss:

   - which groups of trips appear consistently across all three methods (for example, do airport trips from Question 5 form their own group?);
   - what the linear PCA projection reveals or misses compared to the non-linear methods;
   - how well each method preserves local vs. global structure, and whether the distances between clusters and the cluster sizes in each plot can be interpreted;
   - the runtime and sensitivity to hyperparameters and random seed of each method;
   - how well the clusters from the three methods agree. Quantify this with the adjusted Rand index (`sklearn.metrics.adjusted_rand_score`) between each pair of methods, and discuss whether clusters found in a t-SNE or UMAP embedding can be trusted as real groups in the original data.

   Conclude with which method you would recommend for exploring this dataset, and why. [4 marks]

The following code might be useful (but edit it to suit your own needs):

```
from sklearn.decomposition import PCA
from sklearn.manifold import TSNE
import umap

pca = PCA().fit(X_scaled)
```

```
emb_pca = pca.transform(X_scaled)[:, :2]
emb_tsne = TSNE(n_components=2, perplexity=30, init="pca",
                random_state=42).fit_transform(X_scaled)
emb_umap = umap.UMAP(n_neighbors=15, min_dist=0.1,
                     random_state=42).fit_transform(X_scaled)
```

Submission

Work by yourself or in groups of up to four people. Submit your work to Moodle. Your submission should be a PDF and a Jupyter notebook. You should use NeurIPS latex styles as a format of your PDF

Total Marks Available: 63 Marks.

Deadline: 28 October 2026